

Homework: Unconstrained Optimization with Two Variables

GSE 510

Due: Next Class

Instructions: Answer all questions carefully. Show your work. For optimization problems, clearly compute the first-order conditions, second-order conditions, and interpret your answer when appropriate.

Practice Problems: Work together in class

1. Consider the function

$$f(x, y) = 10 + 4x + 6y - x^2 - y^2.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point by solving
$$f_x = 0, \quad f_y = 0.$$
- (c) Compute the second partial derivatives f_{xx} , f_{yy} , and f_{xy} .
- (d) Write down the Hessian matrix.
- (e) Is the critical point a local maximum, local minimum, or saddle point?

2. Consider the function

$$f(x, y) = x^2 + y^2 - 4x - 6y + 20.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Use the second-order condition to classify the critical point.
- (e) What is the minimum value of the function?

3. Consider the function

$$f(x, y) = xy.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Compute the determinant of the Hessian.
- (e) Classify the critical point.

4. Suppose a firm's profit depends on two choice variables, labor L and capital K :

$$\pi(L, K) = 100L + 120K - 2L^2 - 3K^2 - LK.$$

- (a) Compute π_L and π_K .

(b) Solve the first-order conditions:

$$\pi_L = 0, \quad \pi_K = 0.$$

(c) Compute the Hessian matrix.

(d) Use the Hessian to determine whether the critical point is a local maximum.

(e) Interpret the optimal values of L and K .

5. Consider the function

$$f(x, y) = x^2 - y^2.$$

(a) Compute f_x and f_y .

(b) Find the critical point.

(c) Compute the Hessian matrix.

(d) Classify the critical point.

(e) Explain intuitively why this function has a saddle point.

Homework Problems: Submit individually

6. Consider the function

$$f(x, y) = 24x + 18y - 3x^2 - 2y^2.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Use the second-order condition to classify the critical point.
- (e) Find the maximum or minimum value of the function, depending on your classification.

7. Consider the function

$$f(x, y) = x^2 + 2y^2 + 2xy - 6x - 10y.$$

- (a) Compute f_x and f_y .
- (b) Solve the system of first-order conditions.
- (c) Compute the Hessian matrix.
- (d) Compute the determinant of the Hessian.
- (e) Classify the critical point.
- (f) Find the value of the function at the critical point.

8. Suppose a firm chooses advertising A and research spending R . Its profit is

$$\pi(A, R) = 50A + 80R - A^2 - 2R^2 - AR.$$

- (a) Compute π_A and π_R .
- (b) Solve the first-order conditions for the optimal choices A^* and R^* .
- (c) Compute the Hessian matrix.
- (d) Does the Hessian imply that the critical point is a local maximum? Explain.
- (e) Interpret the cross-partial derivative π_{AR} .

9. Consider the function

$$f(x, y) = x^3 - 3x + y^2.$$

- (a) Compute f_x and f_y .
- (b) Find all critical points.
- (c) Compute the Hessian matrix.
- (d) Classify each critical point as a local maximum, local minimum, or saddle point.
- (e) Explain why checking only the first-order conditions is not enough.

10. **Slightly harder problem.** Suppose a firm chooses two products, q_1 and q_2 . Its profit function is

$$\pi(q_1, q_2) = 40q_1 + 50q_2 - 2q_1^2 - 3q_2^2 - q_1q_2.$$

- (a) Compute the marginal profit with respect to each product:

$$\pi_{q_1}, \quad \pi_{q_2}.$$

- (b) Solve the first-order conditions to find the critical point.
- (c) Compute the Hessian matrix.
- (d) Use the Hessian to determine whether the critical point is a local maximum.
- (e) Find the firm's maximum profit.
- (f) Interpret the cross-partial derivative $\pi_{q_1 q_2}$. What does it suggest about the interaction between producing product 1 and product 2?

Reminder: For a twice differentiable function $f(x, y)$, the Hessian matrix is

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

For a critical point:

- If $|H| > 0$ and $f_{xx} > 0$, the point is a local minimum.
- If $|H| > 0$ and $f_{xx} < 0$, the point is a local maximum.
- If $|H| \leq 0$, the point is neither max nor min.